

Exploring Prosodic Prominence Control in Synthesized Golden Speaker's Speech for Pronunciation Training

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1. INTRODUCTION

Computer-assisted pronunciation training uses text-to-speech (TTS) to provide pronunciation input and feedback

Zero-shot TTS (ZS-TTS) can synthesize a learner's voice with native-like pronunciation, creating a **golden speaker** (GS) version of the learner's voice

Self-imitation pronunciation practice: learners imitate the GS with their own voice and native-like pronunciation
→ **Pronunciation differences become more salient**

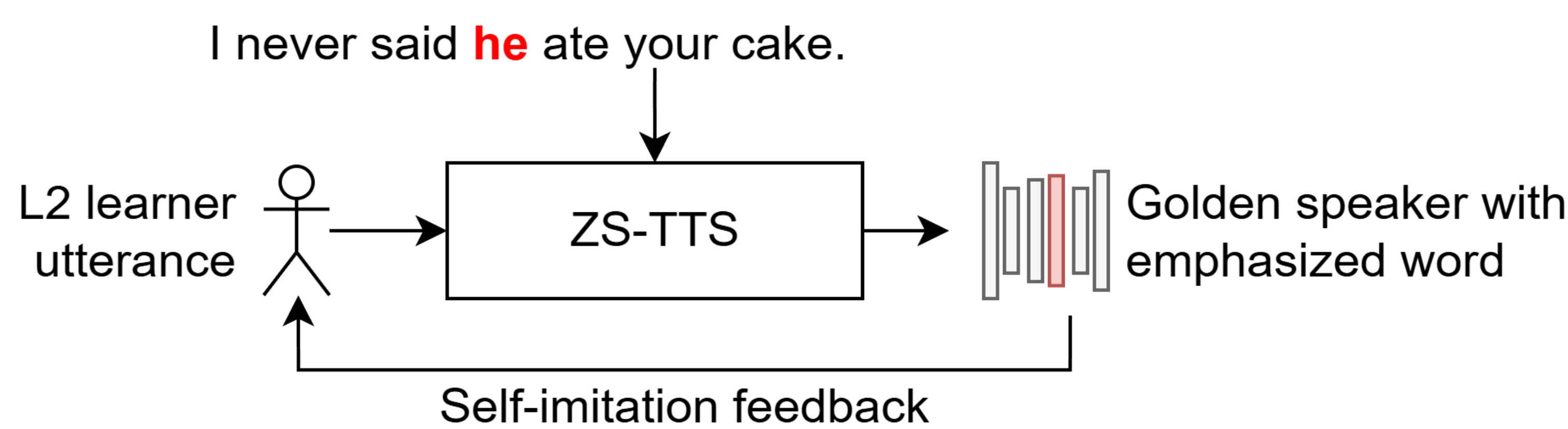
Our previous experiment, university students trained using GS

- Students showed increased interest and comparable pronunciation improvement
- Each sentence contained only one prosodic pattern

To extend this practice, **word emphasis** control using ZS-TTS can be explored

- Recent TTS can follow textual instructions to generate more expressive speech
- Different situations may require emphasizing different words

GOAL: investigate if ZS-TTS can control sentence-level prominence for self-imitation practice with GS



2. EXPERIMENT

ZS-TTS model: CosyVoice 3 [2]

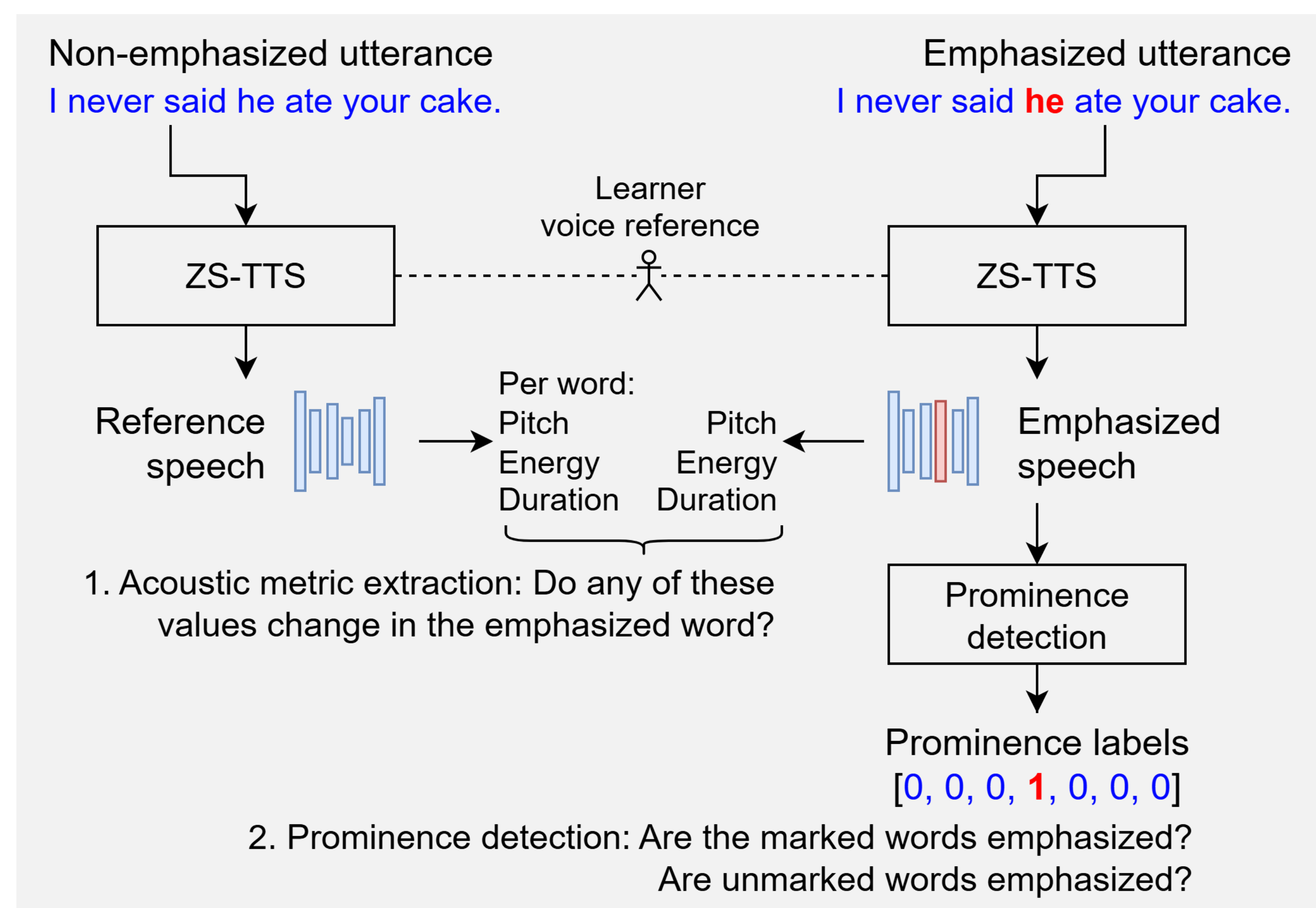
- Example: I never said he ate your cake.
- This input is expected to generate speech in which the word “he” is more prominent

Utterances synthesized with and without emphasis control

- Non-emphasized (neutral), emphasized utterance

A total of 218 sentences from the StressTest [1] dataset

- 4 non-native speakers were used as voice references



Two evaluations were conducted:

1. Acoustic metrics extraction
2. Automatic word prominence detection: WhiStress [3]

3. RESULTS: Acoustic metrics

Question 1 (Q1): How often did the marked word show higher values in the emphasized utterance?

Question 2 (Q2): How often did the marked word show the largest metric increase among all words in the sentence?

	Number of sentences	Metric	Q1 (%)	Q2 (%)
Recordings (Real human speech)	327	Energy	252 (77.1)	152 (46.5)
		Pitch mean	215 (65.7)	100 (30.6)
		Pitch peak	247 (75.5)	123 (37.6)
		Duration ratio	316 (96.6)	226 (69.1)
CosyVoice 3 (Generated)	218	Energy	149 (63.5)	56 (25.7)
		Pitch mean	155 (71.1)	63 (28.9)
		Pitch peak	160 (73.4)	62 (28.4)
		Duration ratio	175 (80.3)	90 (41.3)

- Among the evaluated metrics, duration was the strongest cue for word prominence
- Model increased prominence-related acoustic cues, but with less intensity and less distinctly than human speech
→ Explicit **emphasis instructions produced weaker acoustic changes** than observed in human speech

4. RESULTS: Prominence labels

WhiStress performance with StressTest human recordings

		Human recording annotation	
		Emphasized	Non-emphasized
Predicted label	Emphasized	239 (88.2%)	31 (2.6%)
	Non-emphasized	32 (11.8%)	1142 (97.4%)
F1 score (↑)		0.88	

- F1 score: high values indicate high agreement between predicted and expected prominence labels
→ Although most data points are non-emphasized words, the model achieved robust detection scores

Prominence detection results with synthesized speech

		Instruction to emphasize	
		Yes (%)	No (%)
Predicted label	Total words	1084 (100)	4692 (100)
	Emphasized	600 (55.4)	582 (12.4)
	Non-emphasized	484 (44.6)	4110 (87.6)
F1 score (↑)		0.53	

- 44.6% of marked words were not detected as emphasized
- Even without explicit instructions, 12.4% of unmarked words were detected as emphasized
- Lower F1 score suggests that the generated prominence patterns differ from those observed in human speech
→ **The model struggles to follow explicit emphasis instructions**

5. CONCLUSION

Although the model responded to emphasis instructions to some extent, the findings suggest that it still struggles to generate human-like prominence patterns reliably

References

- [1] Yosha, I. (2025). StressTest: Can YOUR Speech LM Handle the Stress?.
- [2] Du, Z. (2025). CosyVoice 3: Towards In-the-wild Speech Generation via Scaling-up and Post-training.
- [3] Yosha, I. (2025). WhiStress: Enriching Transcriptions with Sentence Stress Detection.

Speech samples

